CCN-NAV — Computer Communication Networks (CS-327) — Master Index

Weeks 1–15 · 17 Aug – 29 Nov 2026 · **Midterm 30** / **Final 40** / **Lab 20** · load 3.5 · difficulty 3.5/5

> **Open this file first.** It is the map: where every topic lives, every line offset, every resource. **Four-file pack** (all in `terms/2026-fall/CCN/`): | File | Holds | |---|---| | `CCN-NAV.md` | This map — glance tables, master offsets, resources index | | `CCN-BEFORE-MID.md` | Weeks 1–8 (midterm) — full per-week content | | `CCN-AFTER-MID.md` | Weeks 9–15 (final) — full per-week content | | `CCN-LABS.md` | All 14 labs + Appendix A + lab viva — lab→week map, per-lab blocks, viva prep |

Where to look (fast answers)

I want…	Open
A topic with its defs/formulas/numericals	that week's block in BEFORE-MID / AFTER-MID
Any week's exact line offsets	**Master offset table** below
A full fear-killer pack (verbatim)	that week's block — packs carry per-question offsets
A lab, step-by-step	`CCN-LABS.md` → per-lab block
Viva questions	`CCN-LABS.md` viva section · `Viva-Book.md`
Books / YT playlists / MOOCs / practice banks	**Resources** section in this file
Exam-day order	**Exam stacks** section
Which official module covers what	**Syllabus modules** section

Course facts

- **Code:** CS-327 · **Credits:** 3+1 (theory + lab) · **Contact:** 3 lectures + 1 lab/week · **Prereqs:** Digital Logic Design, basic programming
- **Weightage:** Midterm 30% / Final 40% / Lab ~20% — **editorial, no official CS-327 grading policy confirmed** (`01-Course-Overview.md` L14–19; `Marks-Allocation.md` agrees). NED Final 60 / Sessional 40 may apply course-wide (`Shared/MASTER-DECISIONS.md` Tier 10). Both kept, labeled.
- **Difficulty:** 3.5/5 · **GPA Risk:** Medium
- **Rotation:** Course B (Wed) + Course A (Thu) — 1 A-slot (Thu, P0 floor anchored) + 1 B-slot (Wed) per week; no Saturday lab (term.json Sat `null`; W8 also Sat `off` for midterm)
- **Midterm:** Week 8 (05–11 Oct) · **Final:** Week 15 (23–29 Nov) · **Lab viva:** Week 15 (separate slot)
- **Exam character: 35% numericals (subnetting, CRC, CSMA/CD, TCP congestion) + 65% theory** (narrative L11). *"Subnetting is the gatekeeper"* — the 30-second /16–/30 target decides A vs A+.
- **Examinability note:** the official syllabus has **no Application Layer** — HTTP/HTTPS/DNS/SMTP/FTP/DHCP/socket programming are **NOT examinable** (`02-Official-Syllabus.md` L89–94). Lab-only labs: 3 (FTP), 13 (Wireshark), 14 (DHCP).

Master weekwise line-offset table

Offsets are line numbers into the named source file. `Fear-Killer-Packs.md` and `Week-by-Week-Narrative.md` are the two anchored sources; `weeks/CCN-Wn.md` files live in `weeks/`. CCN packs are **topic-keyed** (a pack can span multiple manifest weeks — see `weeks/README.md`). W7, W8, W14 have no pack.

Week	Narrative	Pack heading	Pack questions	weeks/CCN-Wn	Module
W1	L17–35	L11 (W1: topic-osi-and-tcpip)	res L12 · Q1 L14 · Q2 L15 · Q3 L16	CCN-W1 L1–57	M1–M3
W2	L39–54	L18 (W2: topic-data-link-layer-and-error-control)	res L19 · Q3 L23	CCN-W2 L1–56	M4

W3	L58–76	L18 (same pack — Q1 CRC, Q2 Hamming)	res L19 · Q1 L21 · Q2 L22	CCN-W3 L1–60	M5
W4	L80–95	L25 (W3: topic-mac-and-csmacd)	res L26 · Q1 L28 · Q2 L29 · Q3 L30	CCN-W4 L1–58	M4
W5	L99–114	L32 (W4: topic-network-layer-and-ip)	res L33 · Q2 L36 · Q4 L38 · Q5 L39	CCN-W5 L1–58	M6–M7
W6	L118–136	L32 (same pack — Q1 VLSM, Q3 subnet)	res L33 · Q1 L35 · Q3 L37	CCN-W6 L1–59	M7
W7	L140–157	— (revision, no pack)	—	CCN-W7 L1–60	M1–M7
W8	L161–171	— (no pack)	—	CCN-W8 L1–31	MIDTERM
W9	L174–194	L41 (W5: topic-routing)	res L42 · Q1 L44 · Q2 L45 · Q3 L46	CCN-W9 L1–60	M8
W10	L198–215	L41 (W5 Q3) + L32 (W4 Q4 IPv6)	res L42 · Q3 L46 · res L33 · Q4 L38	CCN-W10 L1–56	M8 + M7
W11	L219–236	L48 (W6: topic-transport-layer)	res L49 · Q1 L51 · Q2 L52 · Q3 L53	CCN-W11 L1–56	M11
W12	L240–261	L55 (W7: topic-tcp-congestion-control)	res L56 · Q1 L58 · Q2 L59 · Q3 L60	CCN-W12 L1–61	M11
W13	L265–282	L62 (W8: topic-advanced-mpls-sdn-wireless-multimedia)	res L63 · Q1 L65 · Q2 L66 · Q3 L67 · Q4 L68 · Q5 L69	CCN-W13 L1–57	M9, M10, M12, M13
W14	L286–305	— (taper, no pack)	—	CCN-W14 L1–67	all
W15	L307–320	— (no pack)	—	(W15 note at CCN-W14 L66)	FINAL + viva

Syllabus modules → where it lives

CCN has no theory `03-Chapter-Breakdowns/` (only `03-Lab-Breakdowns/` for labs). Module coverage is **pack-level + narrative only** — every row marked ⚠ .

Module	Hours	Breakdown?	Covered in
M1: Introduction to CN	3	⚠ pack-level	W1
M2: OSI & TCP/IP reference models	3	⚠ pack-level	W1
M3: Packet & Circuit Switching	3	⚠ pack-level	W1
M4: Data Link Layer & Issues	4	⚠ pack-level	W2, W4
M5: Error Correction & Congestion Control	4	⚠ pack-level	W3
M6: Network Layer — Protocols & Services	3	⚠ pack-level	W5
M7: IPv4/IPv6, Addressing & Subnetting	7	⚠ pack-level	W5, W6, W10
M8: Network Routing	5	⚠ pack-level	W9, W10
M9: MPLS	2	⚠ pack-level	W13
M10: Wireless Networks	3	⚠ pack-level	W13
M11: Transport Layer — TCP & UDP	5	⚠ pack-level	W11, W12
M12: SDN & VNF	3	⚠ pack-level	W13
M13: Multimedia Networking & Streaming	2	⚠ pack-level	W13

Gaps are flagged, not fabricated: no theory chapter breakdowns exist for CCN — module coverage is verified from `02-Official-Syllabus.md` and embedded pack/narrative content only.

Exam probability table (editorial — study prioritization only)

Verbatim from `01-Course-Overview.md` "Exam Weight Breakdown" (editorial estimates, not official).

Topic	Midterm	Final	Numerical	Diagram	Theory	Definition
OSI/TCP model	90%	40%	0%	30%	45%	25%
DLL/MAC	70%	40%	10%	20%	45%	25%
CRC/Error detection	80%	50%	50%	10%	25%	15%
CSMA/CD	70%	40%	30%	15%	35%	20%
IPv4/NAT	80%	50%	10%	25%	45%	20%
Subnetting	65%	85%	60%	10%	20%	10%
Routing (OSPF/BGP)	0%	75%	25%	20%	40%	15%
TCP/UDP	0%	80%	15%	30%	35%	20%
TCP congestion	0%	75%	25%	25%	35%	15%
IPv6	0%	50%	10%	25%	45%	20%
MPLS	0%	35%	5%	15%	55%	25%
SDN	0%	40%	5%	25%	45%	25%
VNF	0%	30%	0%	15%	60%	25%
Wireless	0%	40%	15%	20%	45%	20%
Multimedia	0%	30%	10%	15%	50%	25%

GPA priority (source: `01-Course-Overview.md`): Subnetting/VLSM/CIDR ▥ Must Win (highest numerical weight) · CRC/Hamming ▥ Must Win · TCP congestion ▦ · OSI/TCP model ▦ · Routing (OSPF/BGP) ▦ · CSMA/CD ▩ · Advanced block (MPLS/SDN/VNF/Wireless/Multimedia) ▩.

Marks allocation

Component	Weight	Strategy
Subnetting/VLSM/CIDR	~20%	30-sec target, Magic Number method
CRC/Hamming	~10%	Binary division drill
TCP congestion control	~10%	Sawtooth trace, Tahoe vs Reno
Routing (OSPF/BGP)	~15%	Algorithm tracing
Advanced block (MPLS/SDN/VNF/Wireless/Multimedia)	~10%	Short-note mastery: SDN planes, MPLS labels, 802.11 CSMA/CA
Theory	~15%	Layer-by-layer protocol mapping
Lab	~20%	Packet Tracer — install Week 1

Weight note: `01-Course-Overview.md` and `Marks-Allocation.md` agree on ~20% lab; both are editorial (no official policy). The official NED policy that may apply is Final 60 / Sessional 40 (`Shared/MASTER-DECISIONS.md` Tier 10) — flagged, not resolved.

Exam stacks

- **Midterm (W8, 05–11 Oct):** past paper 60 min → blank page 30 min → error log 20 min. Answer **numericals first** (subnetting, CRC, CSMA/CD), theory second. Sleep 8 h. Ledger frozen.
- **Final (W15, 23–29 Nov):** subnetting → CRC/CSMA/CD/TCP-cwnd numericals → header/diagram draws → theory. **Subnetting is the highest-ROI marks in the paper — do them first.** Sleep 9 h (banked from W12).

- **Viva (W15, separate slot):** `CCN-LABS.md` viva + `Viva-Book.md` — 3-min Packet Tracer topology walkthrough, then 17 technical questions. Know `show ip route`, `show running-config`, `show vlan brief`, `show interfaces`; be ready to subnet on the whiteboard.

Resources

YouTube playlists

Playlist	Channel	Videos	Why
Computer Networks (Complete)	Gate Smashers	~68	OSI, TCP/IP, subnetting, routing, TCP/UDP, SDN, wireless. 10M+ views
Computer Networking Fundamentals	freeCodeCamp.org	1 (12h)	CRC, flow control, IPv4/CIDR, VLSM, TCP congestion, routing algorithms
Computer Networking: A Top-Down Approach	(textbook companion)	~90	All Kurose & Ross chapters explained

Textbooks

- **Kurose & Ross** — *Computer Networking: A Top-Down Approach*, 8th Ed (2020; Pearson). ISBN **978-0-13-668155-7**. The standard textbook — everything: OSI, TCP/IP, routing, IPv4/6, SDN, wireless.
- **Tanenbaum & Wetherall** — *Computer Networks*, 6th Ed (2021; Pearson). ISBN **978-0-13-676405-2**. Classic reference with strong fundamentals.

Free MOOCs

Course	Source	Notes
Computer Networks and Internet Protocol	NPTEL	IIT Kharagpur, 12 weeks
The Bits and Bytes of Computer Networking	Coursera	Google
Computer Networks	MIT OCW	Advanced, with assignments

Problem banks / practice

Resource	Link
subnetipv4.com	https://subnetipv4.com — infinite interactive subnetting drills (secret weapon)
Kurose & Ross Interactive Exercises	https://gaia.cs.umass.edu/kurose_ross/interactive/
LeetCode Networking Problems	https://leetcode.com/problemset/?search=network
InfoTech Ninja: 25 Subnetting Problems	https://infotechninja.com/subnetting-practice-set-25-problems/

Cheat sheets / revision notes

Resource	Link
GeeksforGeeks CN Cheat Sheet	https://www.geeksforgeeks.org/computer-networks/computer-network-cheat-sheet/
LivePhysics OSI/TCP/IP Cheat Sheet	https://livephysics.com/cheat-sheets/computer-science-computer-networks-osi-and-tcpip-layers/
HowToNetwork CCNA OSI/TCP/IP Cheat Sheet (PDF)	https://www.howtonetwork.com/wp-content/uploads/2021/08/CCNA-OSI_TCP_IP-Cheat-Sheet.pdf

Secret weapon

subnetipv4.com — https://subnetipv4.com — infinite interactive subnetting drills. Subnetting is the gatekeeper (30-sec target); this is the fastest way to reflex-level speed.

Where-everything-lives index (source → embedded location)

Source file	~Lines	Embedded in
`01-Course-Overview.md`	53	NAV facts/weights + exam probability + marks allocation
`02-Official-Syllabus.md`	99	NAV module map + examinability note + textbooks
`03-Lab-Breakdowns/…`	~350	CCN-LABS per-lab blocks (14 + Appendix A, verbatim)
`Definition-Book.md`	28	per-week "Definitions" blocks (verbatim, 12 entries)
`Formula-Book.md`	30	per-week "Formulas" blocks (verbatim)
`Numerical-Book.md`	23	per-week "Numericals" numbered types (#22–32)
`Diagram-Book.md`	23	per-week "Diagrams" numbered refs (#1–17)
`Marks-Allocation.md`	14	NAV (this file)
`Resources.md`	45	NAV (this file) + per-week book/video lines
`Fear-Killer-Packs.md`	72	per-week packs (verbatim, with per-question offsets)
`Week-by-Week-Narrative.md`	329	per-week narrative (verbatim)
`Lab-Schedule.md`	29	NAV/CCN-LABS lab→week map
`Lab-Resources.md`	142	CCN-LABS per-lab Resources lines
`Viva-Book.md`	29	CCN-LABS viva section (verbatim)
`CS-318-…-Workbook.md`	61	CCN-LABS course facts + CLO + group sections
`weeks/CCN-W1…W14` + `README`	~840	per-week anchors, P0 floor, drill targets

> `Top-10-Tricky-Concepts.md` + `Top-100-Questions.md` are **deferred** for CCN (decision 2026-08-05) — equivalent coverage lives in `Definition-Book.md`, `Fear-Killer-Packs.md`, and the topical books. `CCN_CS318_lab.pdf` is the workbook PDF source — parsed into `CS-318-…-Workbook.md`.

Generated from the CCN pack (CS-327). Every offset verified against `weeks/README.md`, `Week-by-Week-Narrative.md`, `Fear-Killer-Packs.md`, and the source files themselves.

Schedule at a glance

W	Dates	Variant	File	Topics	Lab
1	17–23 Aug	P2	BEFORE	OSI/TCP model, PDU, switching	Lab 1 (IP config)
2	24–30 Aug	P2	BEFORE	Physical + DLL, flow control	Lab 2 (UTP cables)
3	31 Aug–06 Sep	P2	BEFORE	CRC, Hamming	Lab 3 (file sharing)
4	07–13 Sep	P2	BEFORE	ALOHA, CSMA/CD	Lab 4 (static routes)
5	14–20 Sep	P0	BEFORE	IPv4 header, classful, NAT	Lab 5 (RIP)
6	21–27 Sep	P2	BEFORE	Subnetting, VLSM, CIDR	Lab 6 (OSPF)
7	28 Sep–04 Oct	P1	BEFORE	Midterm revision	Lab 7 (PPP)
8	05–11 Oct	MIDTERM	BEFORE	Exam week — no new material	—
9	12–18 Oct	P1	AFTER	Routing algorithms (recovery)	Labs 8+9 (ACL + STP)
10	19–25 Oct	P2	AFTER	RIP/OSPF/BGP, IPv6	Lab 10 (VLANs)
11	26 Oct–01 Nov	P2	AFTER	TCP & UDP	Lab 11 (NAT)
12	02–08 Nov	P2	AFTER	TCP congestion control	Lab 12 (BGP)
13	09–15 Nov	P1	AFTER	MPLS, SDN, wireless, multimedia	Lab 13 (Wireshark)
14	16–22 Nov	P0	AFTER	Final taper — past paper FIRST	Lab 14 (DHCP)
15	23–29 Nov	FINAL	AFTER	Exam + lab viva — execute	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week. Variants are editorial tier assignments mirroring SE — CCN manifests carry no variant field. Lab column is official (`Lab-Schedule.md`); W8 (Sat `off`) and W15 no lab; **Labs 8+9 merge into W9**.